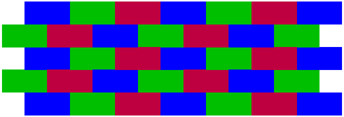
Asymptotic Dimension and Distributed Algorithms

FOCM 17.07.2026

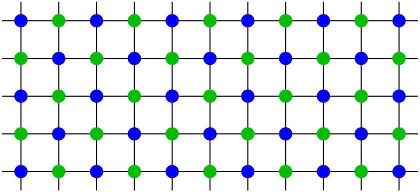
Alexandra Wesolek

Joint work with Marthe Bonamy, Cyril Gavoille
and Timothé Picavet



Graph colouring

k-colouring: Colour **vertices** with k colours such that **vertices** of the same colour are at distance ≥ 2.



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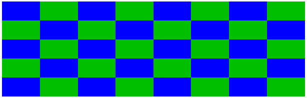

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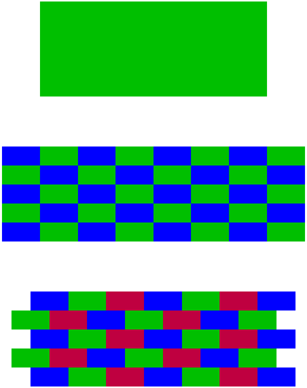

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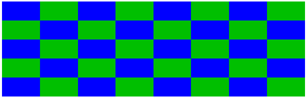

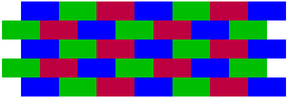

Blocks are coloured with $k = 3$ colours such that **blocks** of the same colour are at distance $\geq r$.

Long story short

If $asdim(X) \leq k$ there exists a $k+1$ colouring of X such that each colour class consists of "small" components which are "sufficiently separated".

f-**block**=set of (weak) diameter $\leq f$

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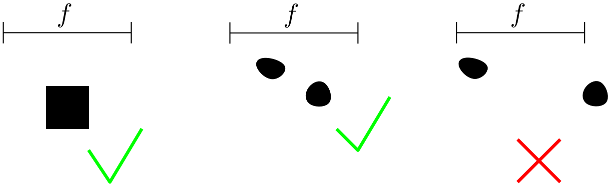

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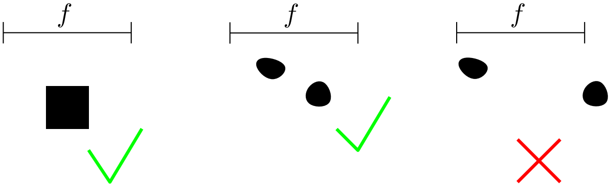

A block can be a disconnected set.

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r-distance k-colouring: For some f, paritioning into f-**blocks** and k-colouring **blocks** s.t. **blocks** of same colour are at distance $\geq r$.

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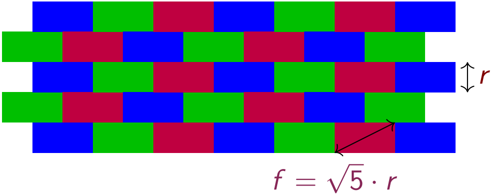

For any r, this is a r-distance 3-colouring.

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Definition

A metric space has <u>*asymptotic dimension*</u> $\leq k$ if for every r it has a r-distance $k+1$-colouring.

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Theorem

The asymptotic dimension of $\mathbb{R}^d$ *is* d.

What is special about *asymptotic* dimension?

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It has nice connections to group theory.

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It can be applied to infinite graphs and graph classes!

History of Asymptotic Dimension

Asymptotic dimension was introduced by Gromov to study finitely generated groups and their Cayley graphs.

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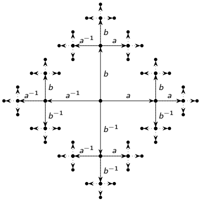

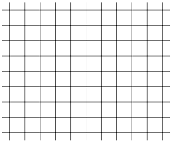

Asymptotic dimension for graph classes $\mathcal{C}$

$\mathcal{C}$ has asymptotic dim $\leq k$ if there exists $f : \mathbb{R}_+ \to \mathbb{R}$ such that: Every graph in $\mathcal{C}$ has an r-distance $k+1$-colouring where blocks are $f(r)$-blocks.

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Theorem (Bonamy, Bousquet, Esperet, Groenland, Liu, Pirot, Scott, 2023)

Let H be a graph. The class of H-minor free graphs has asymptotic dimension at most 2.

Theorem

Trees have asymptotic dimension at most 1.

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$H = C_3$

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Proof.

We define a r-distance 2-colouring as follows (even r):

- Root the tree. Colour first r layers in colour 1, then r layers in colour 2, and so on.

□

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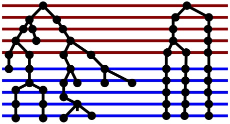
- Collect subtrees in a block which have the same root on level $i - r/2$.



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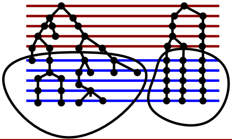
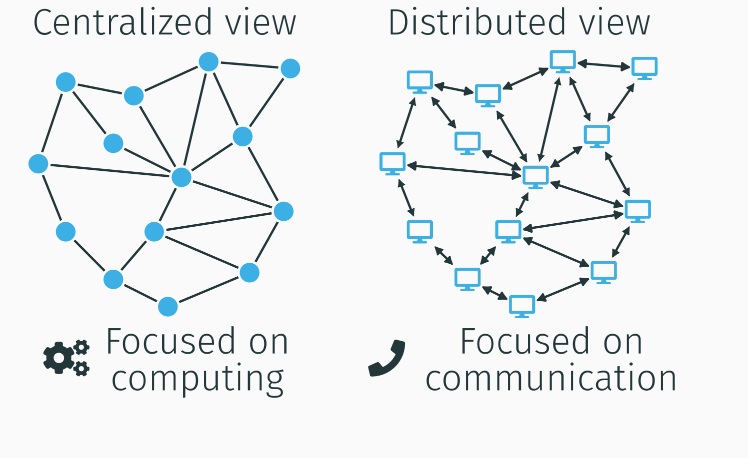
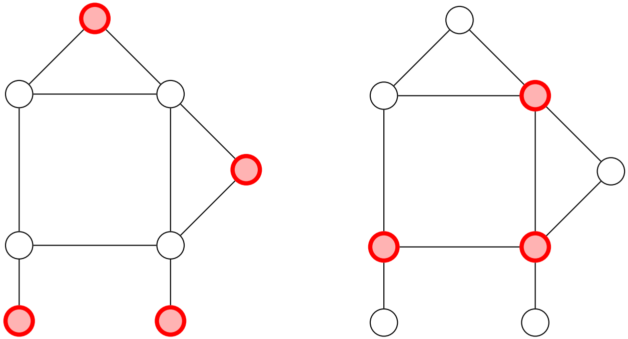


Distributed Algorithms



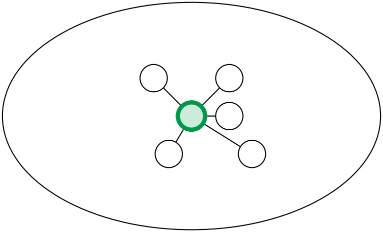
Minimum Dominating Set



$D \subset V(G)$ is a <u>dominating set</u> of G if $\underbrace{D \cup N(D)}_{N[D]} = V(G)$

Complexity in centralized model: NP-complete even on planar graphs

The LOCAL Model



In round 0, each node is only aware of its neighbours.

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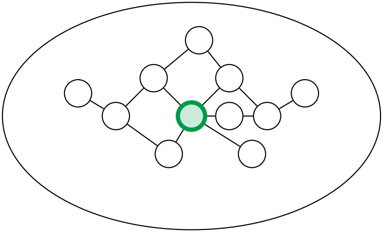

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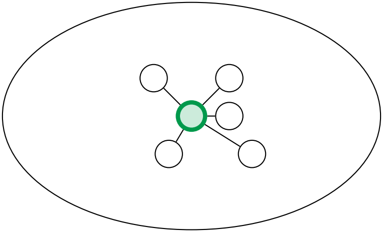

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Complexity of MDS: No approximation algorithm with finite round number
Tractable on planar graphs with 12-approx in 4 rounds

(Bad) Algorithms for forests

Algorithm 1: Optimal Local Algorithm for Forests

Input: A forest G

Result: A set $D \subseteq V(G)$ that dominates G

After n rounds, the highest labelled vertex v of a conn. component H computes min. dom. set D_H of H.

After $2n$ communication rounds, a vertex add itself to D if it is in D_H for some H.

Algorithm 2: Approximation Algorithm for Forests

Input: A forest G

Result: A set $D \subseteq V(G)$ that dominates G

After 0 communication rounds, each vertex v adds itself to the dominating set.

Algorithm 3: 3-Approx Local Algorithm for Forests

Input: A forest G

Result: A set $D \subseteq V(G)$ that dominates G

In round one, every vertex computes its degree and sends it to its neighbors;

$D :=$

$\{\text{Vertices of degree} \geq 2\} \cup \{\text{Vertices with no neighbor of degree} \geq 2\}$;

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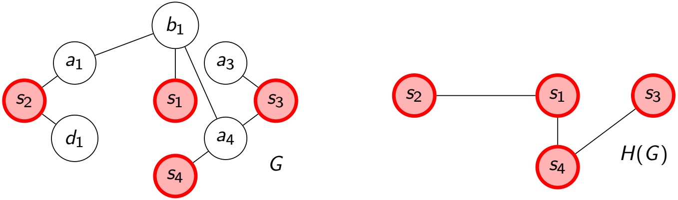

Contract each vertex in $V(G) \setminus S$ to a neighbour in S.

Analysis: $\frac{1}{2}|D \setminus S| \leq |E(H(G))| \leq |S| - 1$

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The algorithm uses 1 round and gives a 3-approximation. Can we give a better approximation when we use 10 rounds? 100 rounds?

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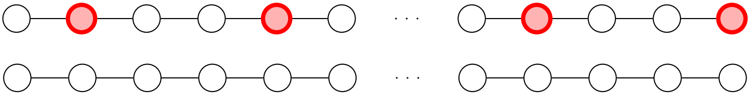

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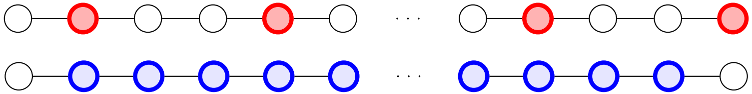
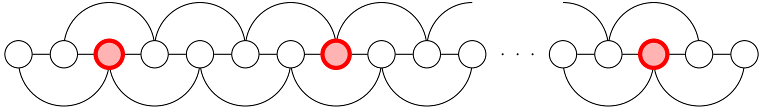


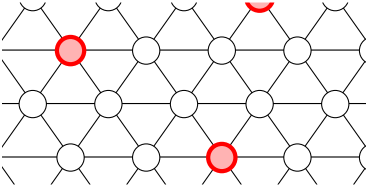
Lower bounds for the LOCAL model

No $(3-\epsilon)$ -approximation for trees possible:

No $(5-\epsilon)$ -approximation for outerplanar graphs possible:



No $(7-\epsilon)$ -approximation for planar graphs possible:



Planar graphs

Theorem (Heydt, Kublenz, Ossona de Mendez, Siebertz, Vigny, 2025)

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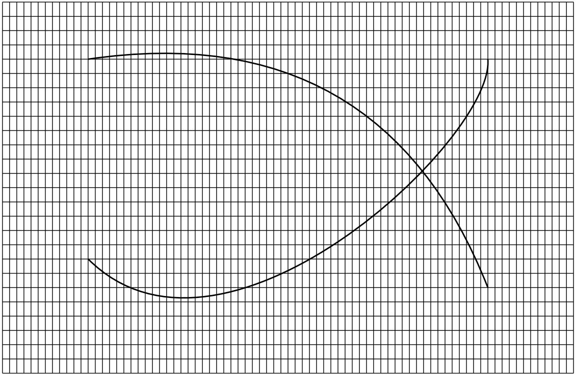
Can be used for **locally planar graphs** with bounded asymptotic dimension.

Suppose we add edges to a planar graph between edges which are far apart.



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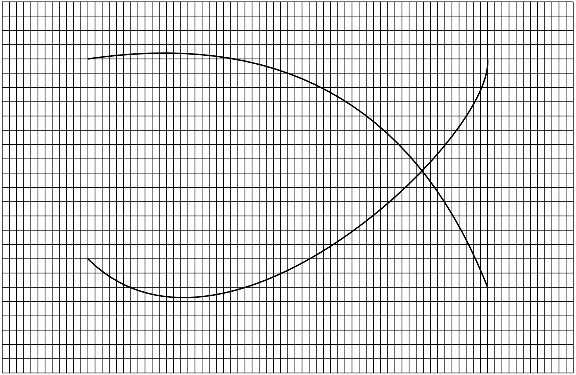

This graph is ‘locally planar’.

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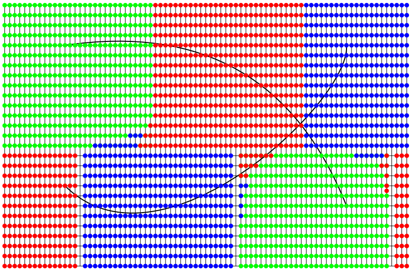
Blocks of size $f(10)$ which are at least 10 apart.



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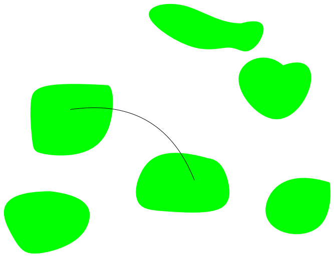

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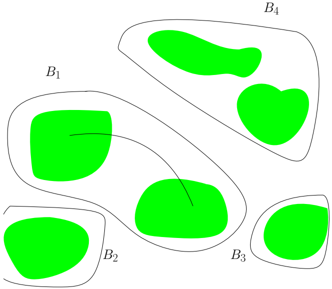

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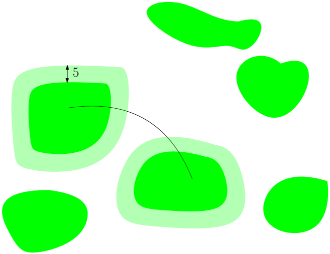

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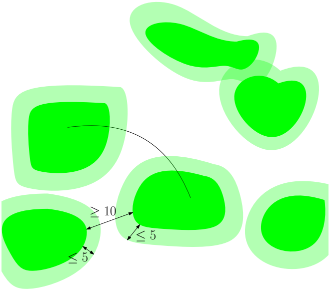

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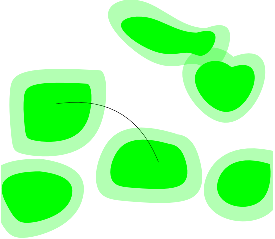
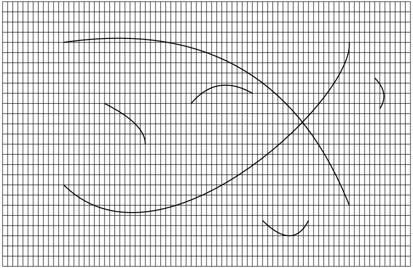


- Take MDS S.
- G_i = 5-th neighbourhood of block B_i
- Delete all vertices in G_i which are not dominated by $G_i \cap S$.
- $d(u)$ for $u \in B_i$ is equal in G, G_i



What if there are also 'local' non-planar subgraphs?

A $\leq$ 2-hop in X is a sequence of vertices $x_1, \ldots, x_k$ in X such that consecutive vertices are of distance ≤ 2 in G.

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Lemma

In a graph of genus g, the 2-hop components of vertices whose ℓ-neighbourhood is not planar have bounded diameter.

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Lemma

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Proof.

Additivity of the genus. $\square$

Algorithm on graphs of genus $\leq g$:

1. If $N^4[u]$ is planar, u runs the 12-approx algorithm.
2. After step 1, brute-force a solution on error set.

The algorithm is $\lesssim 12 \cdot (asdim + 1) = 36$-approx.

Meta-theorems

Additive well-behaved problem
+ k-uniform approx algorithm for $\mathcal{C}$
+ $\mathcal{C}$ hereditary, stable under disjoint union and bdd asim dim
+ locally-$\mathcal{C}$ class $\mathcal{D}$ (up to some errors)
=> approximation algorithm for $\mathcal{D}$

K_t-minor free graphs

Theorem (Kublenz, Siebertz, Vigny, 2021)

There exists a $O_t(1)$-round, $t^{O(t^2 \log(t))}$-approximation LOCAL algorithm for MDS on K_t-minor-free graphs.

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Theorem (Bonamy, Gavoille, Picavet, W., 2025)

There exists a $O_t(1)$-round 50-approximate LOCAL algorithm for Min Dominating Set *on* $\underbrace{K_{2,t}\text{-minor-free graphs.}}_{\text{asdim}=1}$

Open problems:

- Can we get an $O(s)$-approximation for $K_{s,t}$-minor-free graphs?

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- How about if we exclude an interval graph as induced minor/fat minor?
- How can the meta-theorems be used more in practice?